Hands-On-Activity No. 6.2

Title
Built-in Functions

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming Logic and Design	**Date Performed: Oct 22, 2026**
Section: 11S2	**Date Submitted: Oct 22, 2026**
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Objective(s)

This activity aims to develop programming skills using functions.

Intended Learning Outcomes

The students should be able to:
2.1 Understand how to construct programs modularly from small pieces called functions.
2.2 Implement common math functions available in the C standard library
2.3 Create new functions

Materials and Equipment

Internet connection
Personal Computer

Procedure

1. Create a program that defines a function to compute for the area of a cube. The formula of the area of the cube is given as A = s * s * s.
2. Define a function hypotenuse that calculates the length of the hypotenuse of a right triangle when the other two sides are given. Use this function in a program to determine the length of the hypotenuse for each of the following triangles. The function takes two arguments of type double and return the hypotenuse as a double.
3. Implement the following integer functions:
    1. Function celsius returns the Celsius equivalent of a Fahrenheit temperature.
    2. Function fahrenheit returns the Fahrenheit equivalent of a Celsius temperature.
    3. Use these functions to write a program that prints charts showing the Fahrenheit equivalents of a Celsius temperature from 0 to 100 degrees, and the Celsius equivalents of all Fahrenheit temperatures from 32 to 212 degrees. Print the outputs in a neat tabular format that minimizes the number of lines of output while remaining readable.

Output

1. CODE:

```
#include <iostream>
#include <cmath>
using namespace std;

// Function that computes cube area
double computeCube(double s) {
        return s * s * s;
}

int main ()
{
        double s; // Stores side length entered by user

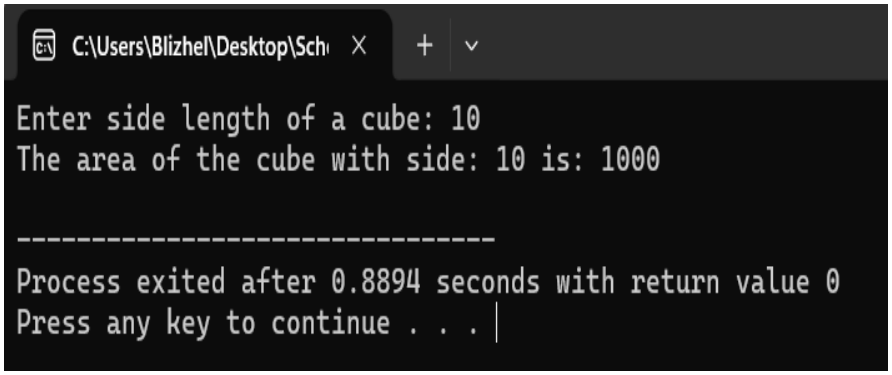
        cout << "Enter side length of a cube: ";
        cin >> s;
```

```
        // Call the function and stores returned value in A
        double A = computeCube(s);

        // Displays the result
        cout << "The area of the cube with side: " << s << " is: " << A << endl;

        return 0;
}
```

OUTPUT:



2. CODE:

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

// Function
double hypotenuse(double s1, double s2){
        return sqrt(s1 * s1 + s2 * s2);
}

int main()
{
        double s1, s2; // Stores two side lengths
        char again; // Stores users choice to continue program or stop

        do
        {
                cout << "Enter side 1: ";
                cin >> s1;

                cout << "Enter side 2: ";
                cin >> s2;

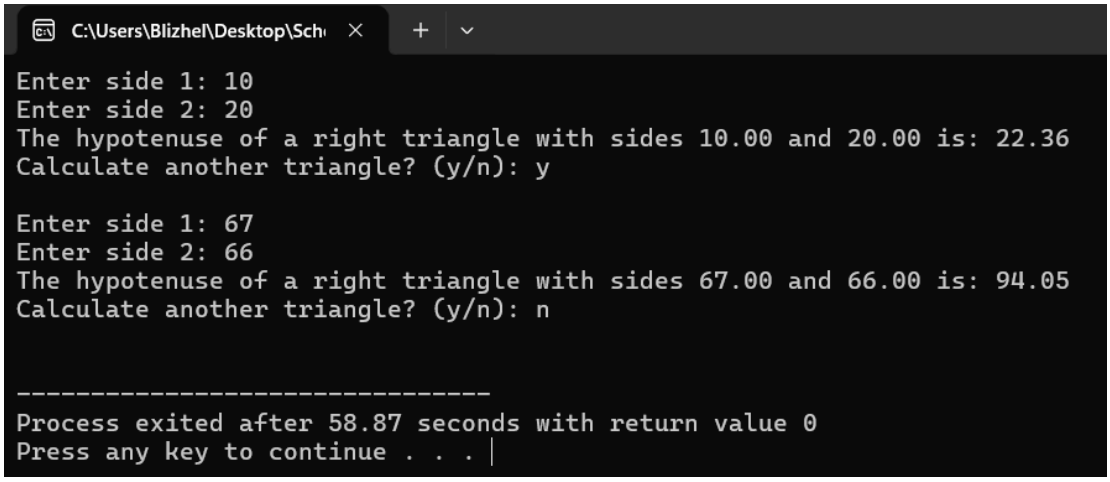
                // Call the function and store value
                double h = hypotenuse(s1, s2);

                // Display result
                cout << fixed << setprecision(2);
                cout << "The hypotenuse of a right triangle with sides " << s1 << " and " << s2 << " is: " <<
h << endl;
```

```
                // Asks user if they want to try again
                cout << "Calculate another triangle? (y/n): ";
                cin >> again;
                cout << endl;
        } while (again == 'y' || again == 'Y');

        return 0;
}
```

OUTPUT:



3. CODE:

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

// Function prototypes
int celsius(int f);
int fahrenheit(int c);

int main()
{
        int count;  // Counts how many pairs have been printed in the row

        // ---------- Chart 1: Celsius 0 to 100 -> Fahrenheit ----------
        cout << "CELSIUS TO FAHRENHEIT" << endl;
        count = 0;
        for (int c = 0; c <= 100; c++)
        {
                cout << setw(4) << c << setw(6) << fahrenheit(c);
                count++;

                if (count == 5)       // Row is full, start new line
                {
                        cout << endl;
                        count = 0;
```

```
                }
                else                // Otherwise add spacing before the next pair
                {
                        cout << "   ";
                }
        }
        cout << endl << endl;

        // ---------- Chart 2: Fahrenheit 32 to 212 -> Celsius ----------
        cout << "FAHRENHEIT TO CELSIUS" << endl;
        count = 0;
        for (int f = 32; f <= 212; f++)
        {
                cout << setw(4) << f << setw(6) << celsius(f);
                count++;

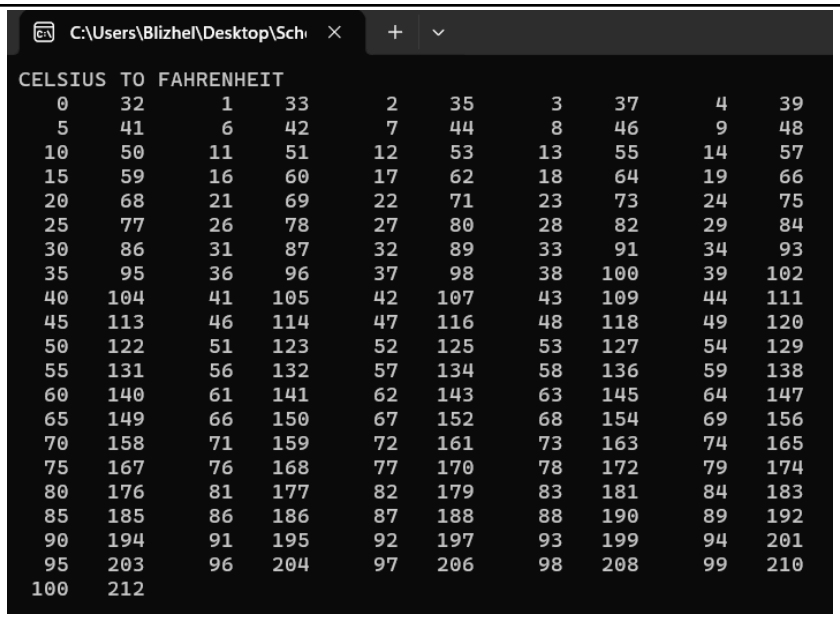
                if (count == 5)
                {
                        cout << endl;
                        count = 0;
                }
                else
                {
                        cout << "   ";
                }
        }
        cout << endl;

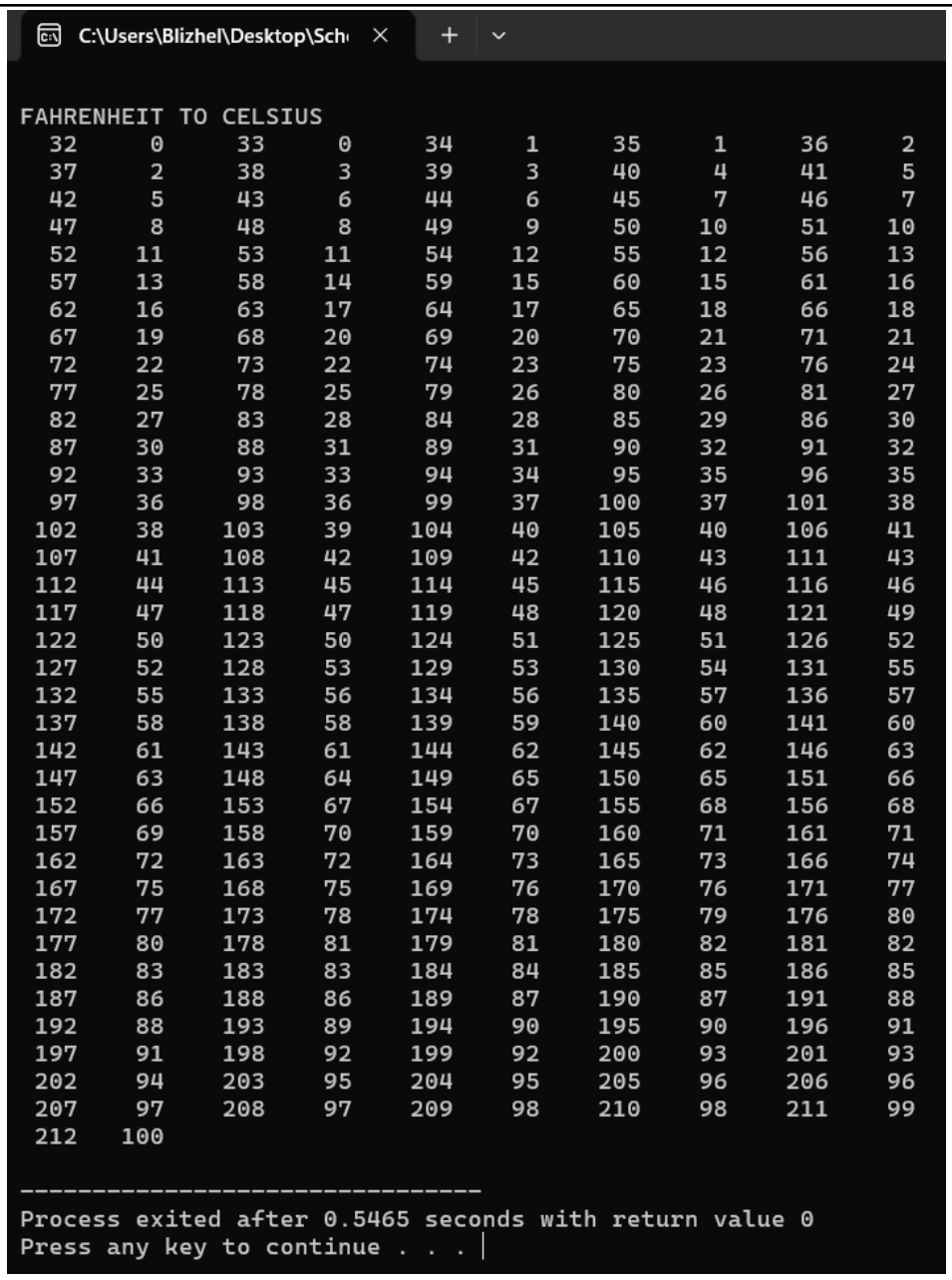
        return 0; // Program ended successfully
}

// Returns the Celsius equivalent of a Fahrenheit temperature
int celsius(int f)
{
        return 5 * (f - 32) / 9;
}

// Returns the Fahrenheit equivalent of a Celsius temperature
int fahrenheit(int c)
{
        return 9 * c / 5 + 32;
}
```

OUTPUT:


C:\Users\Blizhel\Desktop\Sch
CELSIUS TO FAHRENHEIT
0 32 1 33 2 35 3 37 4 39
5 41 6 42 7 44 8 46 9 48
10 50 11 51 12 53 13 55 14 57
15 59 16 60 17 62 18 64 19 66
20 68 21 69 22 71 23 73 24 75
25 77 26 78 27 80 28 82 29 84
30 86 31 87 32 89 33 91 34 93
35 95 36 96 37 98 38 100 39 102
40 104 41 105 42 107 43 109 44 111
45 113 46 114 47 116 48 118 49 120
50 122 51 123 52 125 53 127 54 129
55 131 56 132 57 134 58 136 59 138
60 140 61 141 62 143 63 145 64 147
65 149 66 150 67 152 68 154 69 156
70 158 71 159 72 161 73 163 74 165
75 167 76 168 77 170 78 172 79 174
80 176 81 177 82 179 83 181 84 183
85 185 86 186 87 188 88 190 89 192
90 194 91 195 92 197 93 199 94 201
95 203 96 204 97 206 98 208 99 210
100 212



Conclusion

In this activity, I learned how to solve problems like computing the area of a cube and finding hypotenuse while using functions in C++.